Agricultural production and food security monitor

Staple-crop output, yield and the yield gap, 1990-2022



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Source: Project_07_Agriculture.xlsx (FAO production extract) · 17 survey years, every second year

Start the story

The brief: find where higher yields would add the most food

Aim
Show decision-makers how staple-crop production and yield have moved since 1990, where yields trail the best in each region, and where closing that gap would add the most food.

Questions this report answers

- Is production still growing, and is the growth coming from yield or from more land?
- Which regions and countries produce the most, and at what yield?
- Which crops give the most food per hectare?
- How far is each country below its regional top-quartile yield?
- Where would closing that gap add the most tonnes?

25
countries

11
crops

17
survey years, 1990-2022

7
staple crops

Four FAO regions: Africa (14 countries), Americas (4), SE Asia (4), W Pacific (3).
The brief mentions the Sahel; the data covers four continents, so the report describes it as it is.

Source: Project_07_Agriculture.xlsx · scope of the data · 2

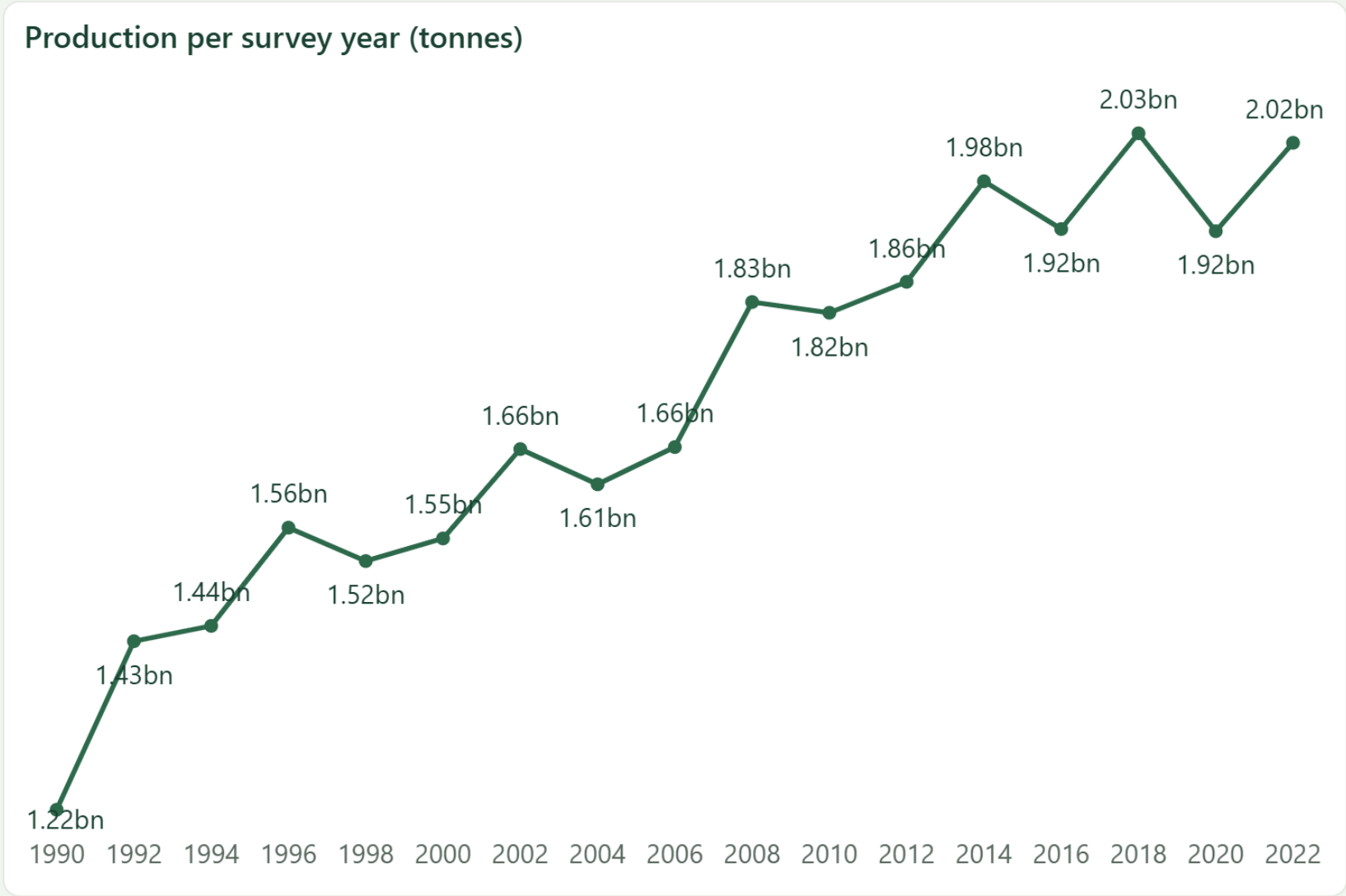
Output grew 65% since 1990, but yield has not moved since 2014

29.04bn	6.40bn	4.54	5.12bn
tonnes produced, all survey years	hectares harvested	average yield, t/ha	extra tonnes if yield gaps closed

STALLED	**Production growth**	Up 61.5% from 1990 to 2014, then only 2.3% more by 2022
FLAT	**Yield**	3.60 t/ha in 1990, 4.98 t/ha in 2014, still 4.98 t/ha in 2022
GAP	**Africa**	Grows 42.8% of all output at the lowest regional yield, 3.67 t/ha
STRENGTH	**Roots and tubers**	Use 14.9% of the farmland but grow 43.9% of the tonnage
OPPORTUNITY	**Closing yield gaps**	18 of 25 countries trail their benchmark; closing it adds 5.1bn t (17.6%)

Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 3

Production rose 61% from 1990 to 2014, then grew only 2% in the next eight years



1.22bn
tonnes in 1990

1.98bn
tonnes in 2014 (+61.5%)

2.02bn
tonnes in 2022 (+2.3% since 2014)

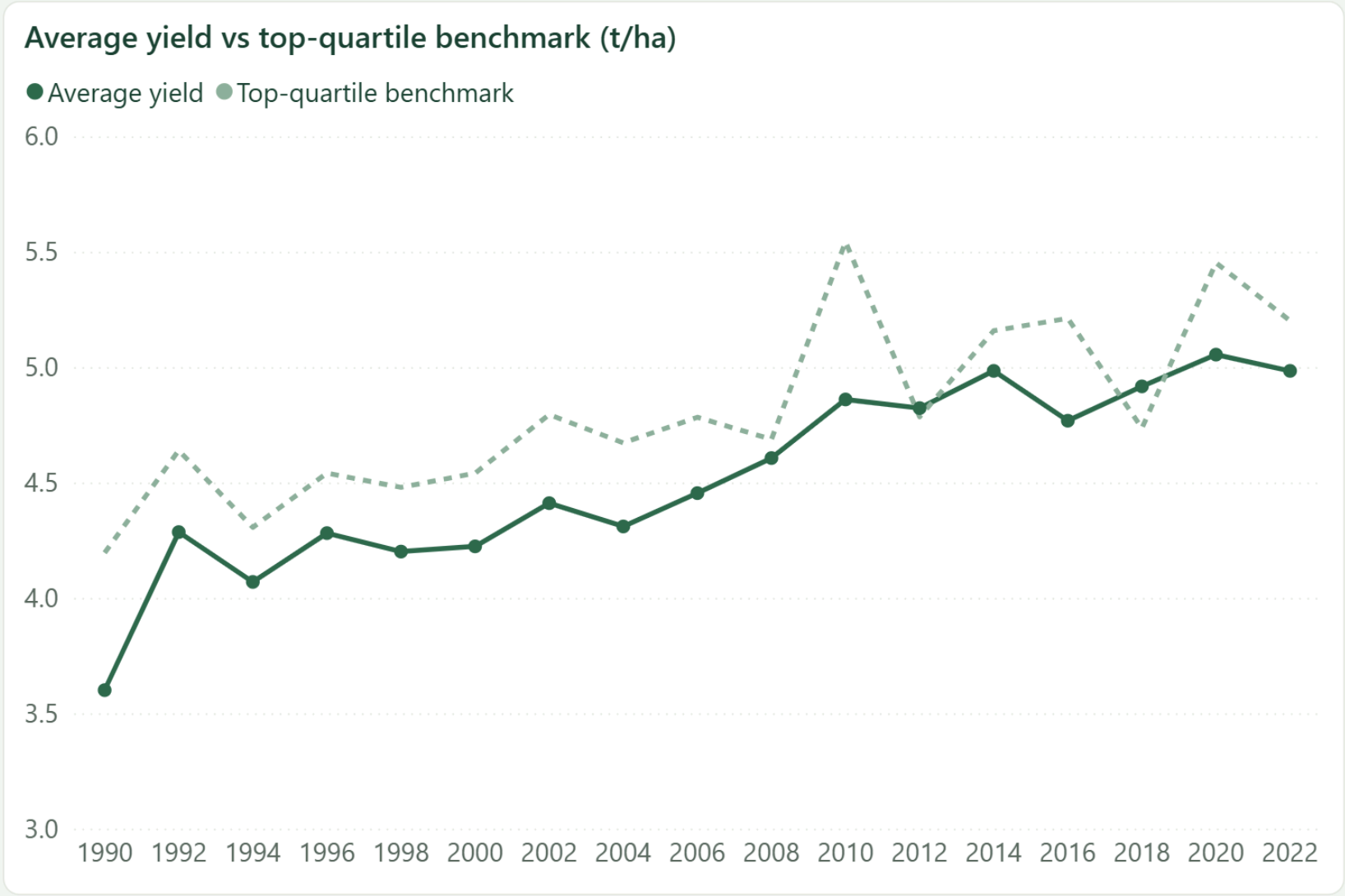
Growth came from yield, not land.

Area rose 17% to 2014 while yield rose 38%. After 2014 both grew about 2%.

Source: Project_07_Agriculture.xlsx · one value per survey year · 4

Dashboards

Yield climbed from 3.60 to 4.98 t/ha by 2014 and has not moved since



3.60
t/ha in 1990

4.98
t/ha in 2014 (+38%)

4.98
t/ha in 2022 (no change)

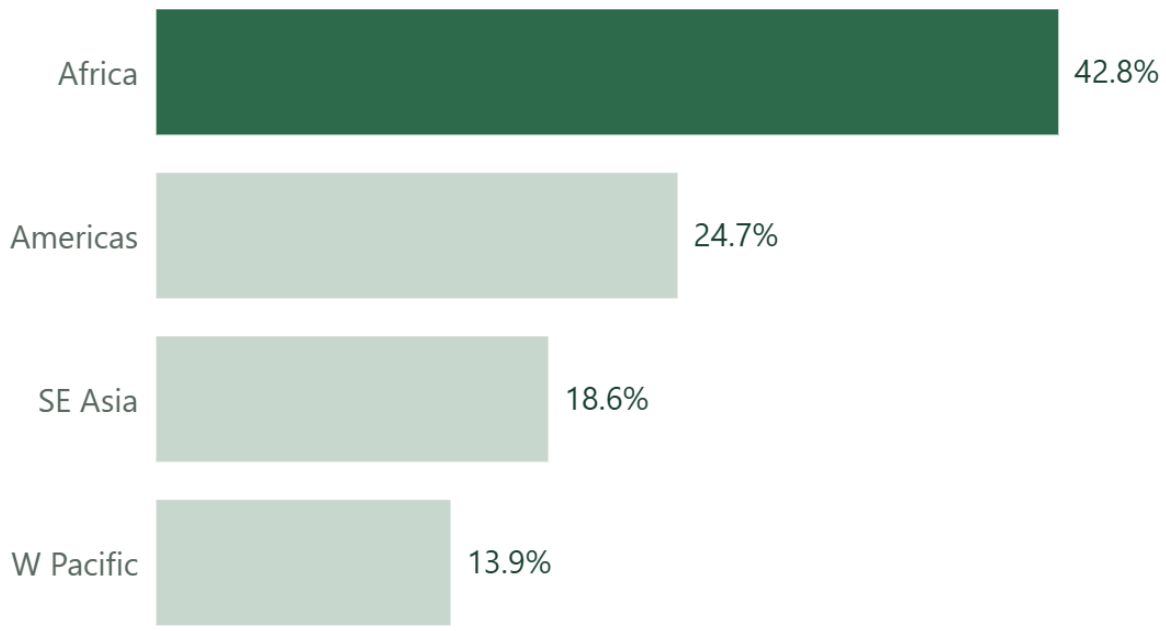
The benchmark is the 75th percentile of country yields.
A country at the benchmark out-yields three in four of its peers.

Source: Project_07_Agriculture.xlsx · yield = production / area harvested, per survey year · 5

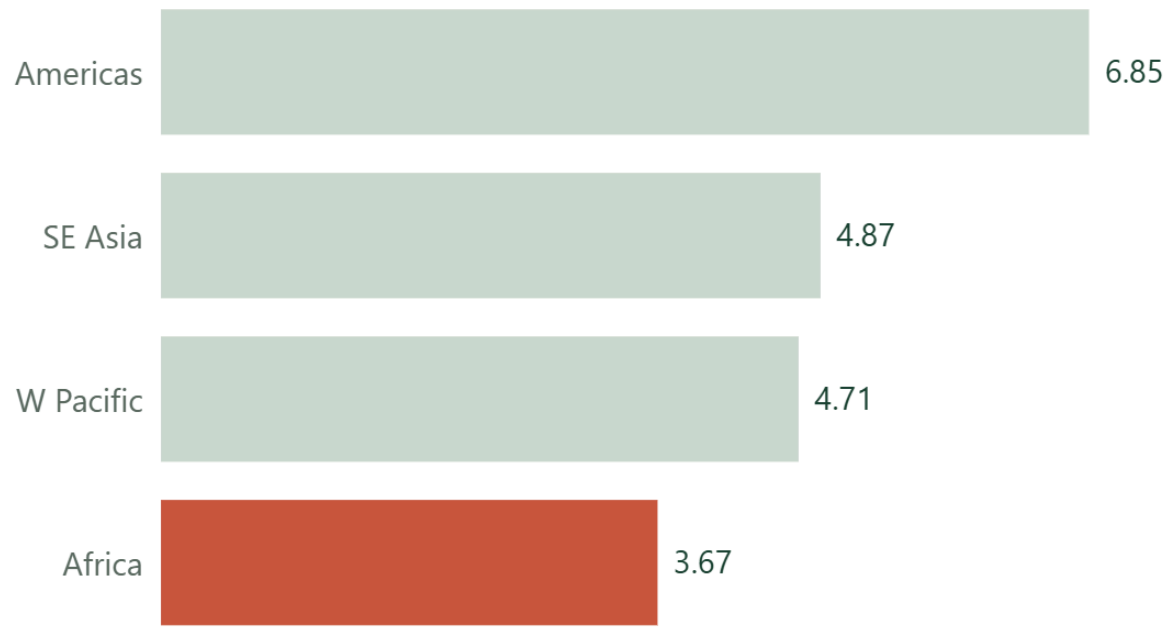
Dashboards

Africa grows 43% of the output at the lowest yield of any region, 3.67 t/ha

Share of production by region



Average yield by region (t/ha)

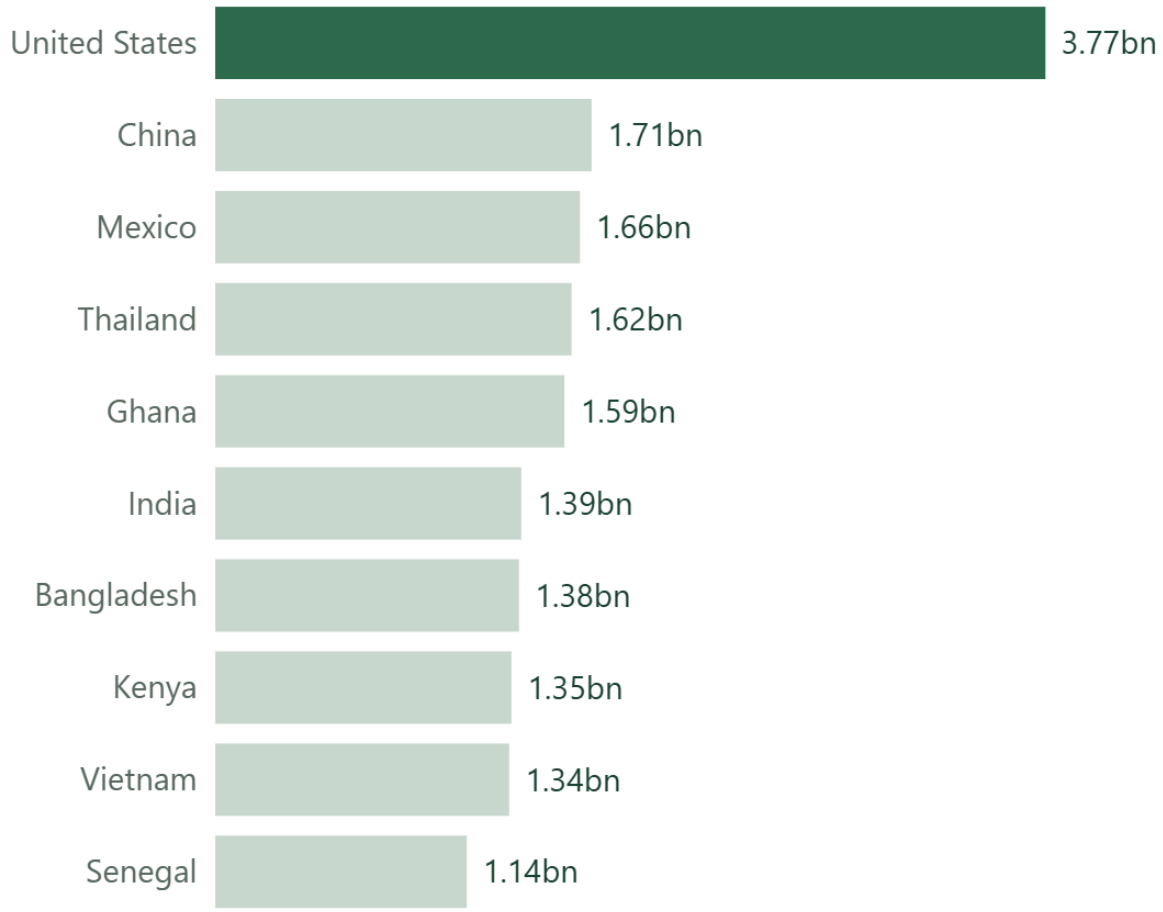


Africa farms the most land at the lowest yield. Raising it to the Americas level would almost double its output.

Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 6

Five countries grow 36% of all output; the United States alone grows 13%

Top 10 producers (tonnes)



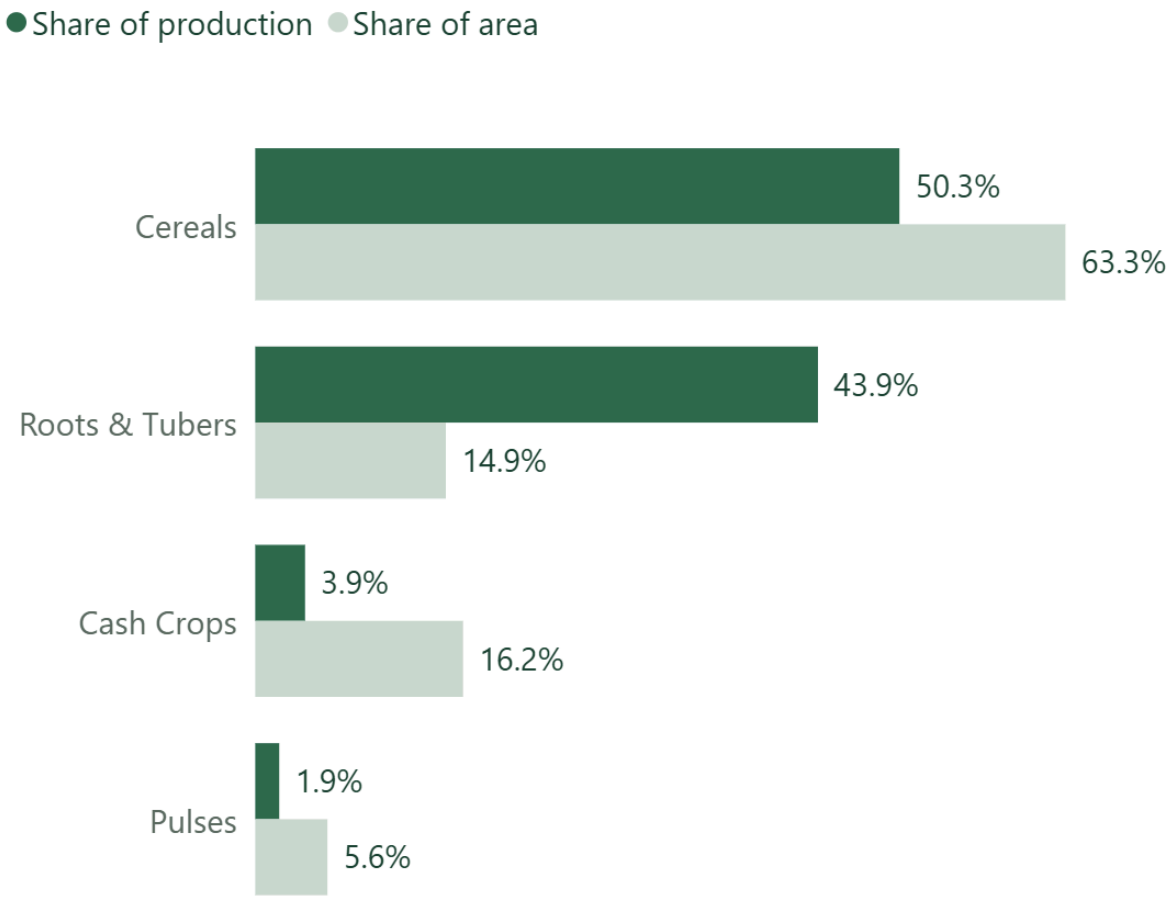
Top 15 countries: production, share and yield

Country	Production (t)	Share	Yield (t/ha)
United States	3,772,522,340	13.0%	10.57
China	1,710,087,526	5.9%	5.89
Mexico	1,657,028,455	5.7%	5.68
Thailand	1,619,121,835	5.6%	7.50
Ghana	1,586,367,213	5.5%	4.15
India	1,390,597,988	4.8%	4.13
Bangladesh	1,380,456,940	4.8%	4.49
Kenya	1,345,703,265	4.6%	4.60
Vietnam	1,335,732,349	4.6%	4.80
Senegal	1,142,952,092	3.9%	4.46
Egypt	1,124,007,762	3.9%	3.49
Indonesia	1,005,669,893	3.5%	4.05
Philippines	1,005,371,963	3.5%	3.45
Brazil	929,106,112	3.2%	4.63
Ethiopia	928,737,646	3.2%	3.53

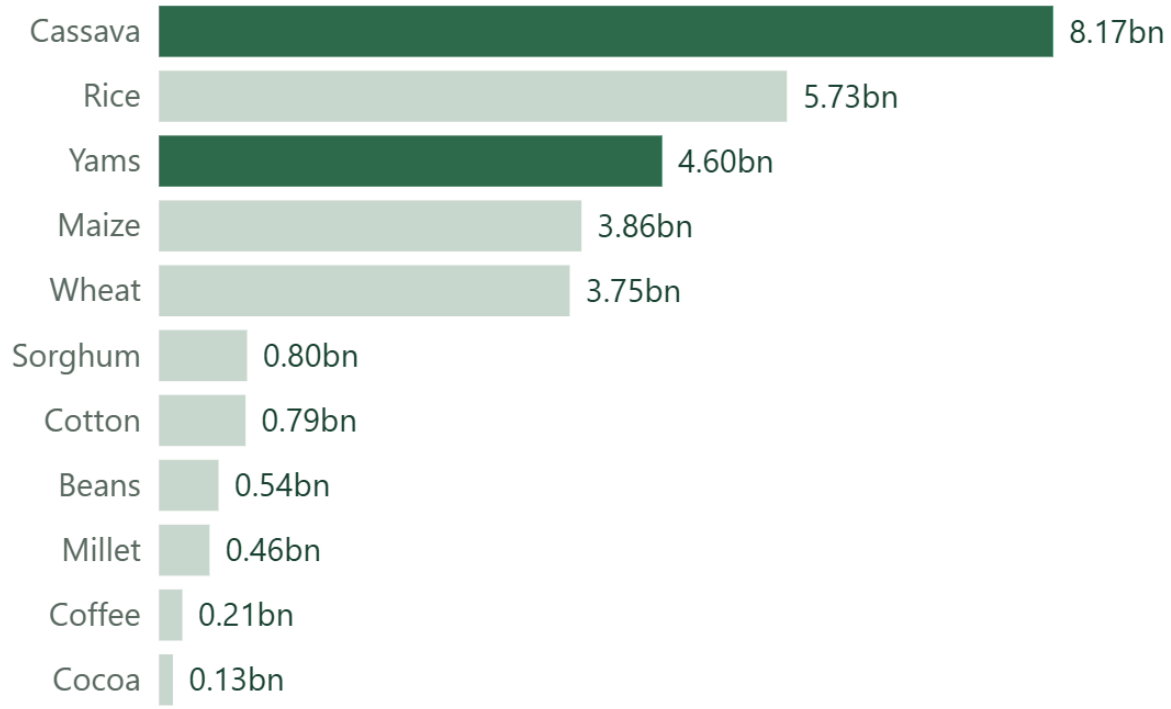
Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 7

Roots and tubers use 15% of the farmland but grow 44% of the tonnage

Share of production vs share of area



Production by crop (tonnes)

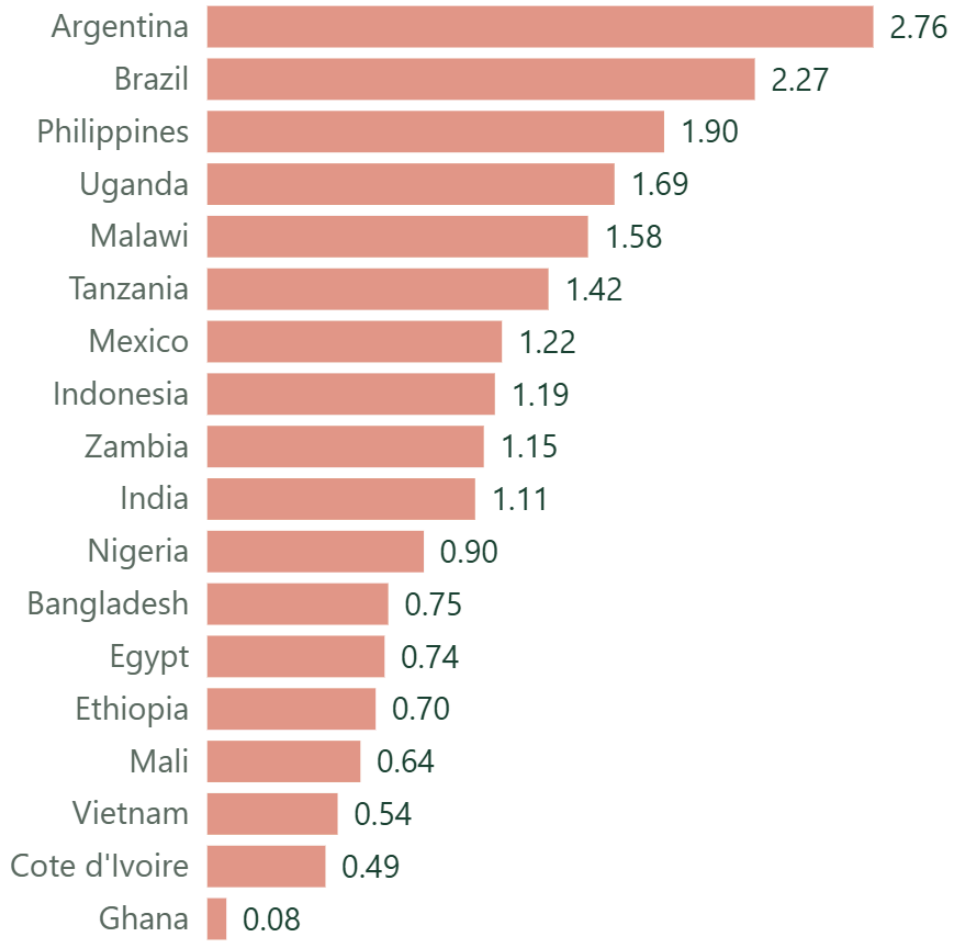


Tonnes are not calories: cassava carries about 1,100 kcal/kg, maize 3,700.

Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 8

18 of 25 countries yield below the top quartile of their own region

Yield gap to regional benchmark (t/ha)



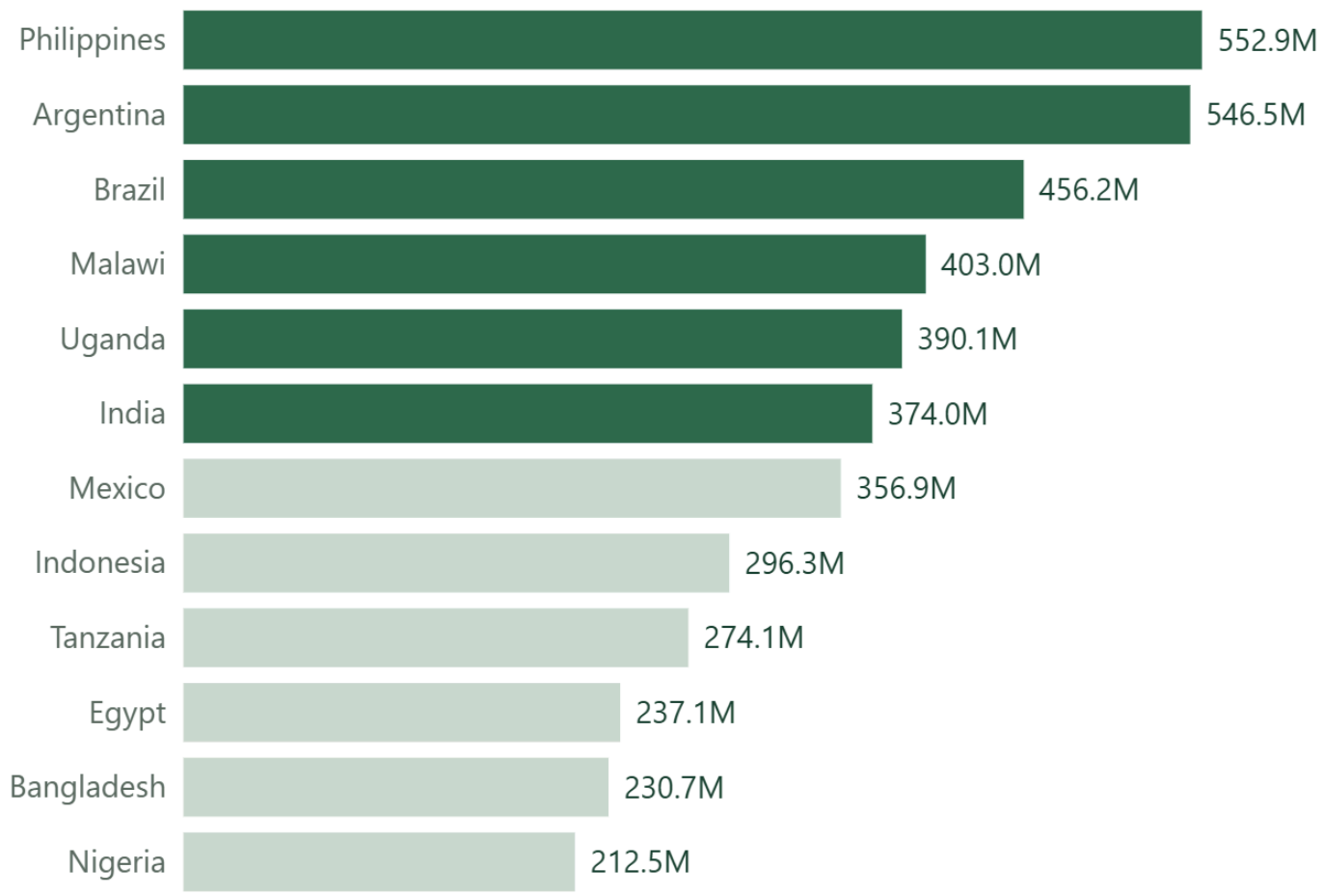
15 largest gaps: yield vs regional benchmark (t/ha)

Country	Region	Yield	Benchmark	Gap
Argentina	Americas	4.14	6.90	2.76
Brazil	Americas	4.63	6.90	2.27
Philippines	W Pacific	3.45	5.34	1.90
Uganda	Africa	2.54	4.23	1.69
Malawi	Africa	2.65	4.23	1.58
Tanzania	Africa	2.81	4.23	1.42
Mexico	Americas	5.68	6.90	1.22
Indonesia	SE Asia	4.05	5.25	1.19
Zambia	Africa	3.08	4.23	1.15
India	SE Asia	4.13	5.25	1.11
Nigeria	Africa	3.33	4.23	0.90
Bangladesh	SE Asia	4.49	5.25	0.75
Egypt	Africa	3.49	4.23	0.74
Ethiopia	Africa	3.53	4.23	0.70
Mali	Africa	3.59	4.23	0.64

Source: Project_07_Agriculture.xlsx · benchmark = 75th percentile of country yields in the region · 9

Closing the gap would add 5.1bn t; six countries hold over half of it

Extra tonnes if each country reached its benchmark



5.12bn
extra tonnes, all survey years

17.6%
on top of actual production

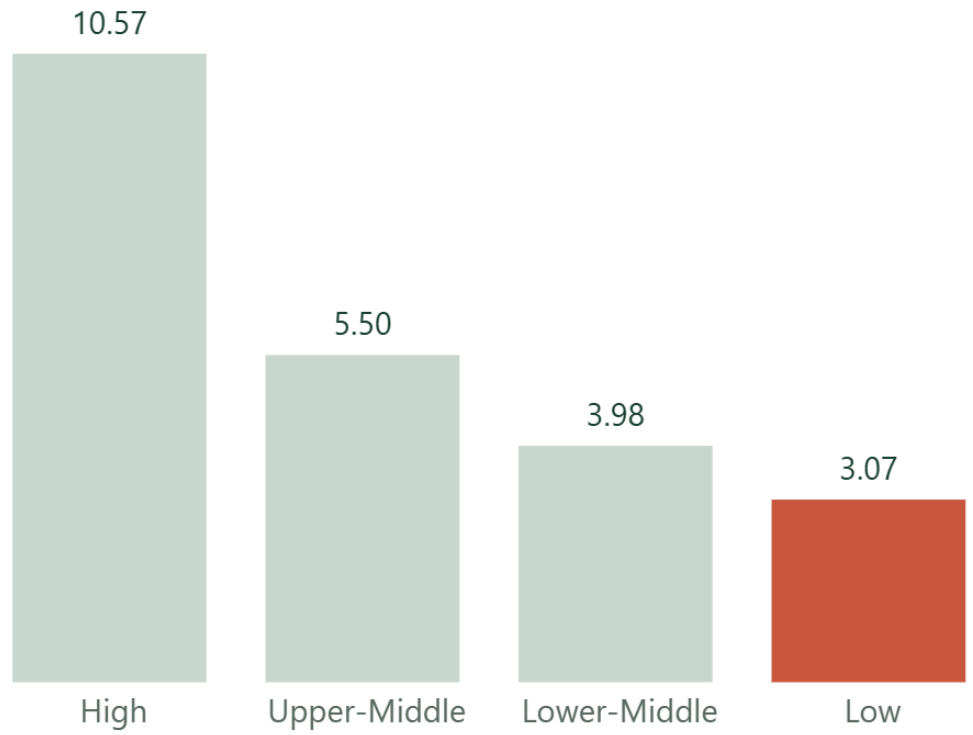
18
countries below benchmark

Gain = yield gap x area harvested.
Large farm areas, not only large gaps, put Argentina and Brazil near the top.

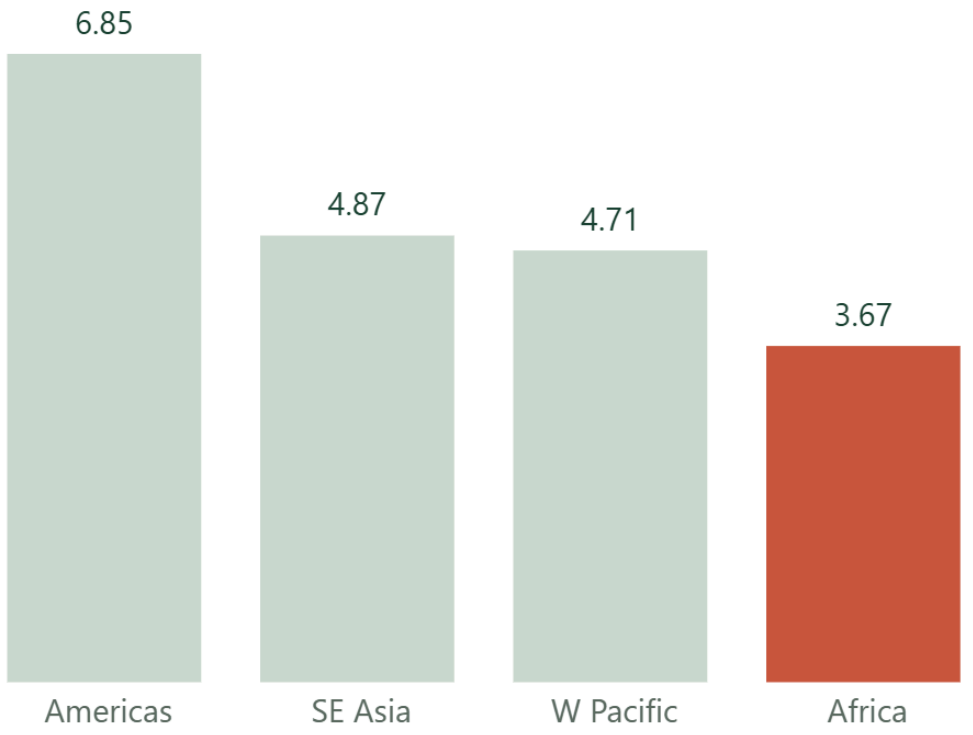
Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 10

High-income farms yield 10.6 t/ha, more than three times low-income farms at 3.1

Average yield by income group (t/ha)



Average yield by region (t/ha)



Read with care: the High group is one country (the United States), and all four Low-income countries are in Africa.

Source: Project_07_Agriculture.xlsx · totals are cumulative over the 17 survey years · 11

Dashboards

Three actions: close Africa's gap, restart yield growth, back roots and tubers

1

Close Africa's yield gap

Seeds, fertiliser and extension in the 14 African countries, where yields trail the regional benchmark of 4.23 t/ha.

Target: Malawi and Uganda alone could add about 0.8bn t

2

Restart yield growth

Yield has been flat at 4.98 t/ha since 2014. Put research and irrigation money into the six largest-gain countries.

Target: yield growth back above 2% per survey round

3

Back roots and tubers

Cassava and yams give the most tonnes per hectare. Fund storage and processing so the harvest reaches markets.

Target: keep their share of output above 40%

5.12bn
extra tonnes at benchmark yield

18
countries below benchmark

17.6%
more food from the same land

Source: Project_07_Agriculture.xlsx · targets are planning goals, not forecasts · 12

Assumptions and data issues behind these numbers

How to read the totals

- Totals add up all 17 survey years (1990-2022, every second year). Trend pages show one year at a time.

Corrections to the original report

- Benchmark: the old measure averaged the top three-quarters of countries; it is now the 75th percentile.
- Potential gain: a divide-by-10,000 hg/ha conversion was removed; yield is already in t/ha.
- Region shares were swapped on the old insights page; Africa is 42.8%, not the Americas.

Data limits

- Population figures are not realistic (China 13M), so per-person measures are not used.
- 252 values are blank (78 area, 84 production, 90 yield) and are left out of the sums.

14.6%

of records are FAO estimates, not official figures

Estimates and coverage by region

Region	Estimated	Countries	Yield (t/ha)
Africa	14.6%	14	3.67
Americas	14.1%	4	6.85
SE Asia	14.5%	4	4.87
W Pacific	15.2%	3	4.71

Estimates are spread evenly.

Every region sits between 14.1% and 15.2%, so they do not tilt the regional comparisons.

Source: Project_07_Agriculture.xlsx · all records · 13

PRODUCTION · 25 COUNTRIES · 11 CROPS · 1990-2022

Output stalled after 2014

Survey year: All | Region: All | Crop: All

29.04bn tonnes produced

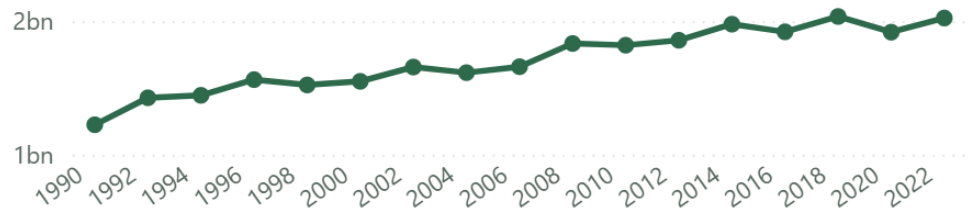
6.40bn hectares harvested

25 countries

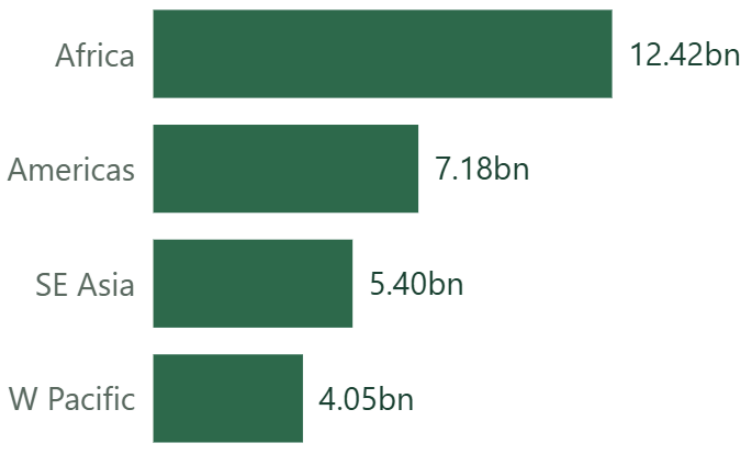
4.54 average yield, t/ha

17 survey years

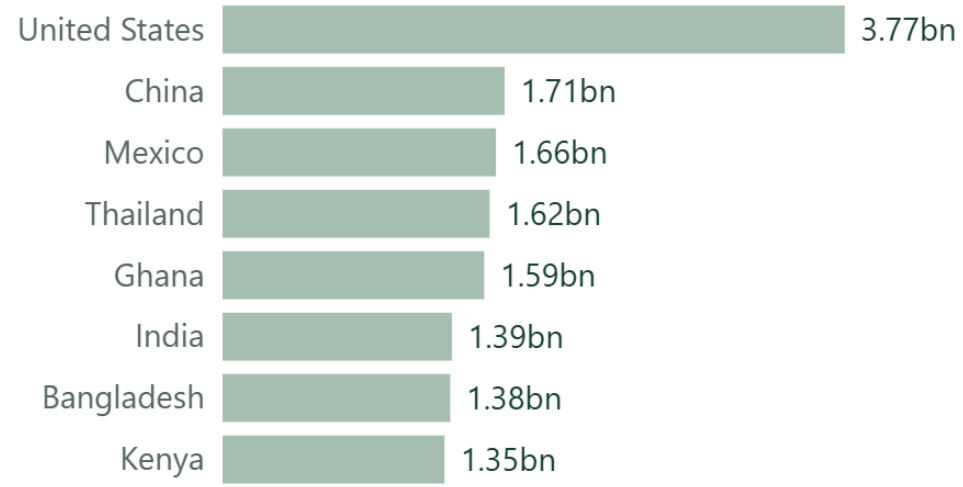
Production per survey year



Production by region



Top 8 producers (right-click for profile)



WHAT THIS SHOWS

Production rose 61.5% to 2014, then only 2.3%.

Africa grows 42.8% of all output.

Five countries grow 36% of it.

Story | **Production** | Crops | Yield gap

Taiwo Ajayi
Report author



CROPS · 25 COUNTRIES · 11 CROPS · 1990-2022

Roots and tubers out-yield cereals

Survey year: All | Region: All | Crop: All

Cassava

largest crop by tonnage

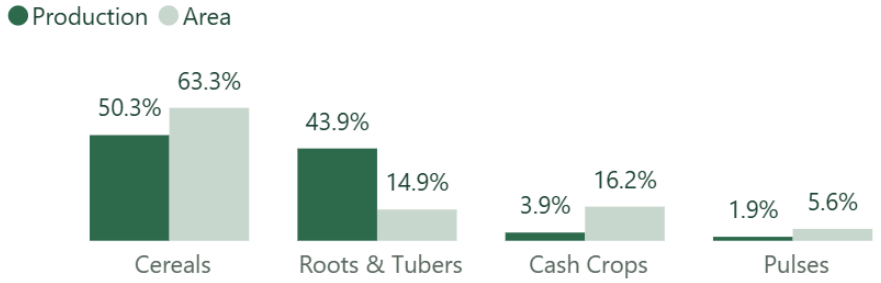
29.04bn tonnes produced

11 crops in view

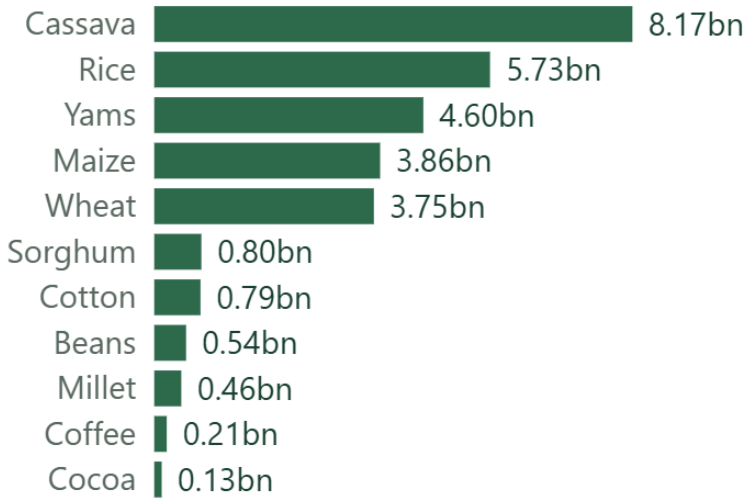
Cereals largest crop family

7 staple crops

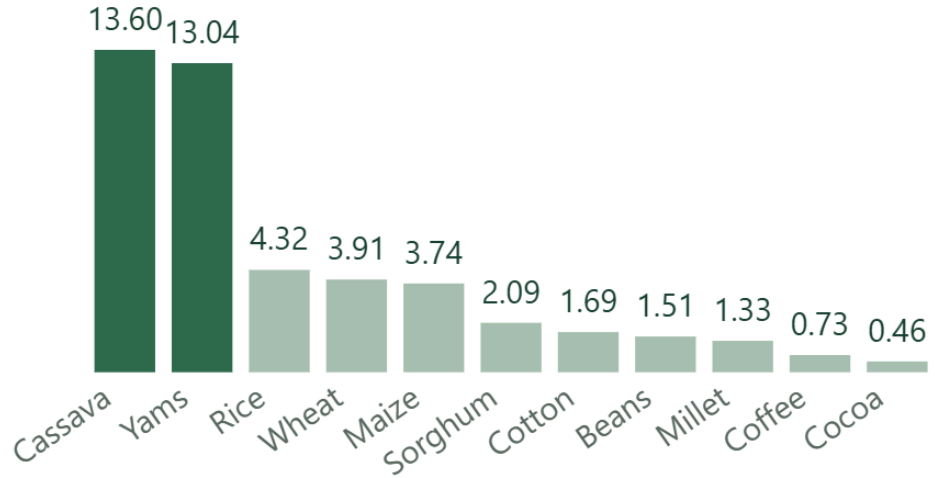
Share of production vs share of area



Production by crop



Average yield by crop (t/ha)



WHAT THIS SHOWS

Cassava and yams yield about 13 t/ha, three times the cereals.

Roots and tubers grow 43.9% of output on 14.9% of the land.

Tonnes are not calories: cassava has a third of maize's energy per kg.

Story | Production | Crops | Yield gap

Taiwo Ajayi
Report author



YIELD GAP · 25 COUNTRIES · 11 CROPS · 1990-2022

Closing the gap adds 5.1bn t

Survey year: All | Region: All | Crop: All

5.12bn
extra tonnes at benchmark yield

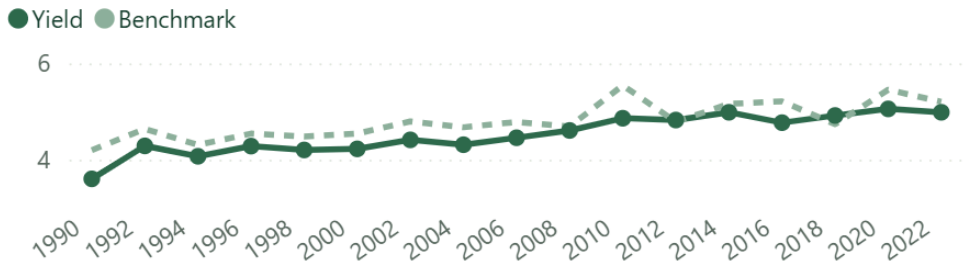
4.54
average yield, t/ha

18
below benchmark

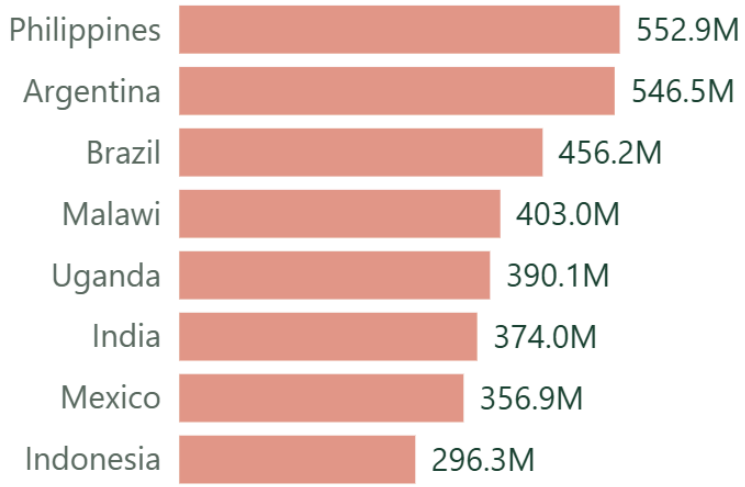
4.63
benchmark, t/ha

17.6%
more food, same land

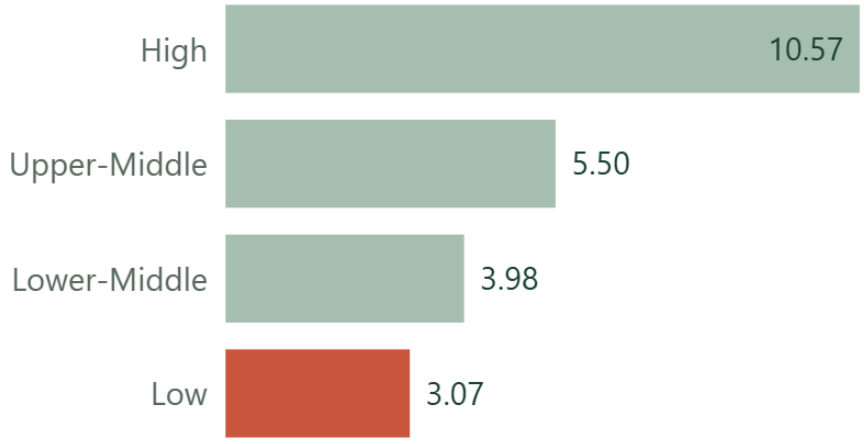
Yield vs benchmark (t/ha)



Potential gain, top 8 (right-click for profile)



Average yield by income group (t/ha)



WHAT THIS SHOWS

18 of 25 countries yield below the top quartile of their region.

Six countries hold over half of the 5.1bn t gain.

Low-income farms yield 3.07 t/ha against 10.57 in the High group.

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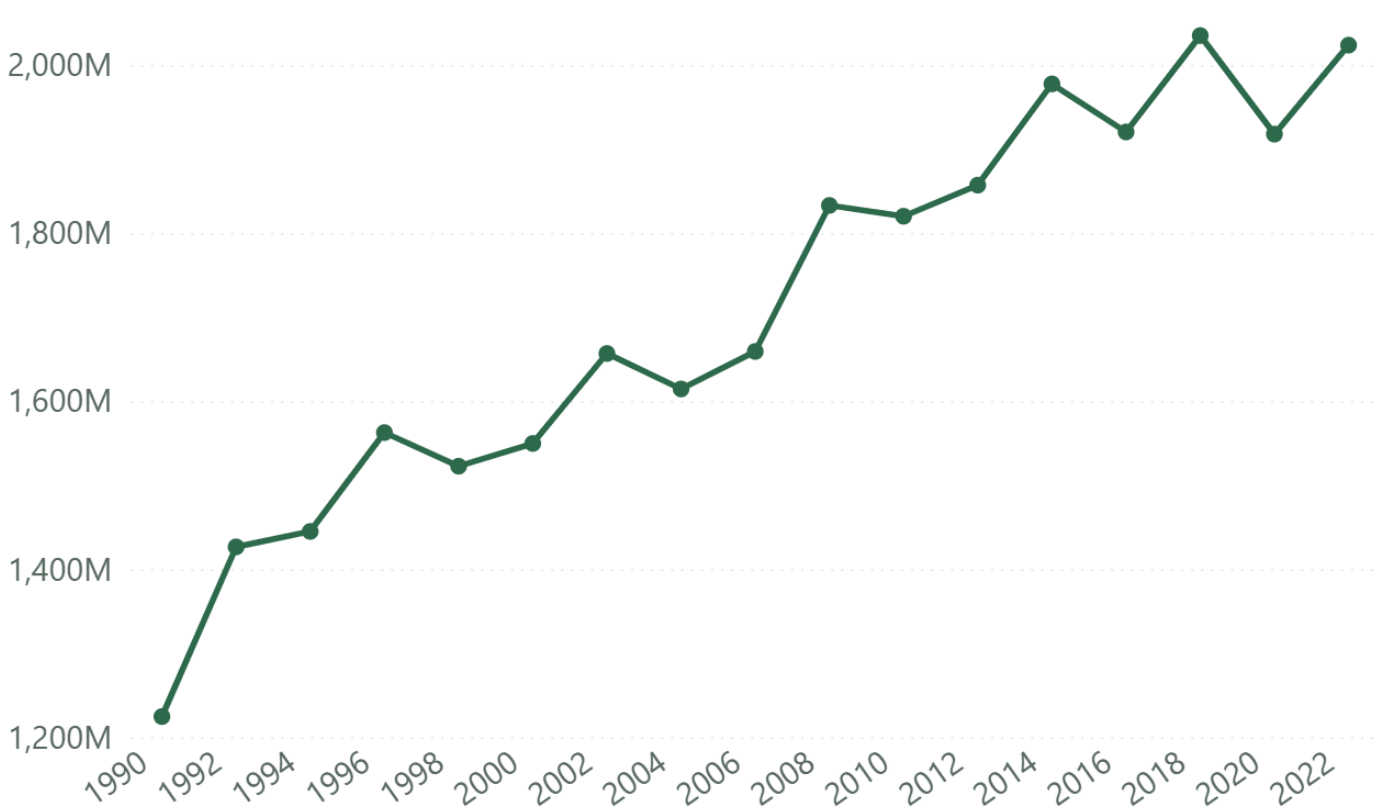


Country profile: production, yield and gap for the selected country

Back

29,044.1M	4.54	4.63	0.09	5,118.0M
Production (t)	Yield (t/ha)	Regional benchmark (t/ha)	Yield gap (t/ha)	Potential extra tonnes

Production per survey year (tonnes)



Production by crop

